

Downloading Vintages from the ECB Real Time Database

Experience Report and Construction of the Vintages

Johns Hopkins University, Department of Economics | October 2026 |  JChaquica/ecb-rtd-vintages

Executive Summary

All 261 vintages from January 2001 to September 2026 can be extracted by following the [RTD page on the ECB Data Portal](#). Many series in the recent vintages are out of date, however, because the ECB stopped updating them. From 2015 to 2020 no series was out of date, although nine had no data at all. In 2023–2025 between 16 and 79 of the 278 series were out of date or empty in a vintage, and in 2026 more than 100 are out of date, with 107 in the latest vintage (Section 3). The vintages from 2015 are available only through the ECB’s web service (its API), which is open and needs no key or account. The instructions for 2001–2014 are clear and accurate, but for later vintages they are not. For the later vintages, the page provides a single example for reference, and several statements on the page are wrong: the update schedule, the description of the pre-2001 files, and the coverage of the web service, which actually starts in January 2001, not 2015. **The main difficulty is that the address returns a log of revisions rather than vintages, and neither the page nor the API help explains how to rebuild a vintage from it.** One series can be downloaded without programming, by pasting the page’s address into a browser; the full set of 278 series takes 278 slow requests and is practical only with a script. See Section 5 for a list of suggestions for the ECB.

1 Website Claims vs. Findings

Topic	Website Says	Found
Update schedule	“updated semi-annually, at the beginning of January and July”	False. The page itself shows “Dataset last update 9 September 2026”, and that vintage was online by 3 October. The meeting of 23 July 2026 has no vintage.
Freezing	“always takes place on a working day ahead of the Governing Council meeting at precise time 3.30 p.m.”	Mostly true. Every row is stamped 15:30, and 248 of the 261 dates are the working day before a monetary policy decision.

Topic	Website Says	Found
Before 2001	A zip file of “38 variables containing data vintages prior to January 2001”, in CSV	False. Only one, producer prices, has vintages before January 2001 (from October 1999); none goes beyond June 2009. The read-me inside the zip refers to a file of cut-off dates, <code>Cut-off.xls</code> , that is not in the zip. And the files in the zip are Excel, not CSV.
2001–2014	Three zip files (monthly , quarterly , annual)	True. Each holds 168 CSV files, one per Monthly Bulletin, each a full snapshot.
From 2015	“can be extracted using ECB Data Portal web services”	True. This works for one series per request. Also, without <code>includeHistory=true</code> the address returns only the latest vintage. It also goes further back than the page says: January 2001 onwards.
Technical documentation	“recommended to become familiar with the technical documentation, prior to proceeding with the data download”	Unclear. Names each zip-file column by letter, but gives no web-service keys and does not mention <code>includeHistory</code> or the columns of the download.

2 How the Vintages Were Built

Step 1: Finding a series and its key

A series is named by a key of five parts, but the page does not list the series the dataset holds. Real GDP, for example, is `Q.Sø.S.G_GDPM_TO_C.E`. Its five parts, separated by dots, are the frequency (`Q`, quarterly), the area (`Sø`, the euro area), the seasonal adjustment (`S`, adjusted), the economic concept, meaning what the series measures (`G_GDPM_TO_C`, gross domestic product in chain-linked volumes), and the unit (`E`, euro).

The address below returns every series, with its description.

<https://data-api.ecb.europa.eu/service/data/RTD?detail=nodata&format=csvdata>

The file holds 278 rows, one per series. Three of them, as an example, look like this.

KEY	TITLE	TITLE_COMPL (shortened)
<code>RTD.Q.Sø.S.G_GDPM_TO_C.E</code>	Gross domestic product at market prices, volume	Gross domestic product at market price – Chain linked volumes – Euro
<code>RTD.Q.Sø.S.L_ULC_CONST.X</code>	Unit Labour Cost (based on persons), deflator	Unit Labour Cost – Construction – Index
<code>RTD.Q.Sø.S.L_ULC_INDTO.X</code>	Unit Labour Cost (based on persons), deflator	Unit Labour Cost – Industry, including energy – Index

The two Unit Labour Cost rows are different series carrying the same `TITLE`, so search `TITLE_COMPL` instead, which spells out what each series is. Where two rows share a `TITLE_COMPL`, they are the

same series with and without seasonal adjustment, which the third part of the key tells apart. The file also gives each part of the key in a column of its own, so series can be filtered by what they measure in the `RT_ECON_CONCEPT` column. Once the series is found, its key is in the `KEY` column.

Step 2: Downloading its log

Request the series on its own, by its full five-part key, from the address

```
https://data-api.ecb.europa.eu/service/data/RTD/<key>?includeHistory=true&format=csvdata&detail=dataonly
```

The `KEY` column gives real GDP as `RTD.Q.Sø.S.G_GDPM_TO_C.E`, but the address already contains `RTD`, so only the part after it goes in. With `RTD.` left on the front of the key, the ECB refuses the request. The address for real GDP, for example, is

```
https://data-api.ecb.europa.eu/service/data/RTD/Q.Sø.S.G_GDPM_TO_C.E?includeHistory=true&format=csvdata&detail=dataonly
```

The option `includeHistory=true` asks for every series version rather than the latest one only; `format=csvdata` gives a CSV file; and `detail=dataonly` leaves out the columns of descriptive attributes, which are not needed to rebuild the vintages.

The address also accepts a key with parts left empty, which fetches several series at once. Such a request silently leaves out every removal, so periods the ECB withdrew would wrongly stay in the rebuilt vintages. Each series has to be requested with its full key.

The server sometimes sends a file cut short while reporting success, so I downloaded each series twice and kept it only if the two copies were identical and the last row was complete.

Step 3: Rebuilding a vintage from the log

The ECB updates the database on particular days, its releases. At each release it adds the newest periods and revises some earlier ones, and the log records each of these as a `Replace` row. Periods that had no revisions get no new row. Occasionally the ECB removes periods altogether, and the log records that as a `Delete` row. Each row gives the period it refers to (`TIME_PERIOD`), what happened to it (`ACTION`), the release on which a new value took effect (`VALID_FROM`) or on which the period was removed (`VALID_TO`), and the value itself (`OBS_VALUE`). A vintage is the series as it stood on a given date, with the euro area's membership and the definitions in force at the time.

Real GDP in the second half of 2005 shows how a vintage is rebuilt, as an example. These are the rows its log holds between August and November 2005 for three of its quarters, in euro millions. They are the oldest quarter, 1991-Q1, which the ECB removed in that time, and the two newest, 2005-Q2 and 2005-Q3, which it added; the quarters in between are left out to keep the table short.

TIME_PERIOD	ACTION	VALID_FROM	VALID_TO	OBS_VALUE
1991-Q1	Replace	3 Aug 2005		1,275,507
1991-Q1	Replace	31 Aug 2005		1,273,486
1991-Q1	Replace	2 Nov 2005		1,275,571
1991-Q1	Delete		30 Nov 2005	
2005-Q2	Replace	31 Aug 2005		1,641,807
2005-Q2	Replace	2 Nov 2005		1,643,676
2005-Q2	Replace	30 Nov 2005		1,783,300
2005-Q3	Replace	30 Nov 2005		1,794,581

To rebuild the vintage of a given date, I looked at each period's rows and kept only those dated on or before that date. Each row carries a single date, the day its new or revised value took effect (**VALID_FROM**) or the day its period was removed (**VALID_TO**). The latest row kept decides the period's value. If it brings a new or revised value (**Replace**), that value (**OBS_VALUE**) is the period's value in the vintage. If it removes the period (**Delete**), the ECB had taken the period out by then, so the vintage has no value for it. If no row was kept, the ECB had not yet published the period, so the vintage has no value for it either.

For the release of 30 November 2005, the latest row for 1991-Q1 is the **Delete** of that day, so the quarter is left out. For 2005-Q2 and 2005-Q3 it is the **Replace** of that day, so the vintage holds 1,783,300 and 1,794,581, the second appearing for the first time. Doing the same for each release gives the vintages.

Release date, 2005	3 Aug	31 Aug	5 Oct	2 Nov	30 Nov
1991-Q1	1,275,507	1,273,486	1,273,486	1,275,571	<i>removed</i>
2005-Q2	<i>not yet</i>	1,641,807	1,641,807	1,643,676	1,783,300
2005-Q3	<i>not yet</i>	<i>not yet</i>	<i>not yet</i>	<i>not yet</i>	1,794,581

Between them the two tables show every case. A quarter is added, as 2005-Q2 on 31 August; a quarter is revised, as 1991-Q1 on 31 August; a figure carries over because no row came in, as on 5 October; and a quarter is removed, as 1991-Q1 on 30 November. The large revision to 2005-Q2 on 30 November was made on the same day the ECB removed 1991–1994.

One case was ambiguous. When a period carries both a **Delete** and a **Replace** on the same day, the rule cannot choose between them, and letting the **Delete** win would make the series appear to lose data. I made the **Replace** win.

Step 4: Building the whole dataset

The vintages are meant to line up, so that every series has a vintage on every release and all 278 can be read as they stood on the same day. That needs the full list of releases, the days on which the ECB updated the database. The ECB publishes no such list, but every date in a log's **VALID_FROM** or **VALID_TO** column is one. No single log holds them all, because a series only has a row on the releases where it changed. Pooling the dates in all 278 logs gives the complete set, 261 releases between January 2001 and September 2026, and every series gets a vintage on each, carrying its figures over on the releases where it did not change.

I then applied the rule from Step 3 to every series at every one of those dates and wrote the result in two forms. One is a workbook per series per vintage, 67,532 in all, in the layout used by ALFRED, the St. Louis Fed's archive of vintages, which suits reading a single vintage. The other is one table per series, with periods in rows and release dates in columns, which suits studying revisions.

The scripts, the results of the checks and the lists of out-of-date series are at the address below.

<https://github.com/JChaquica/ecb-rtd-vintages>

The vintages themselves are too large to keep there and can be rebuilt with the scripts or supplied on request.

Step 5: Checking the result

The vintages were checked in four ways.

The rebuild matches the ECB's own vintages wherever those hold data. For 2001–2014 the ECB published the vintages themselves as zip files, an independent record of what the method should produce. Every series was rebuilt for each of the 168 months and compared with that month's file. Where the file holds data for the series, 42,132 comparisons, the rebuilt vintage has the same periods and values in every one, with no tolerance for rounding.

In a further 296 comparisons, covering 20 series, the file's column is empty but the rebuilt vintage is not. Most are the five government bond yields from February 2008 to March 2011, which the ECB stopped publishing after January 2008 but did not remove from the log until 6 April 2011. The rebuilt vintages keep these values, and `old_file_check.csv` lists every case for a user who wants to drop them.

The later vintages agree with outside sources. The ECB published no vintage files from 2015. Instead, nine quarterly series equal Eurostat's figures for the euro area in every quarter since 1995, and the newest quarter of real GDP carries the growth rate from the Eurostat release just before each vintage date.

The rule for same-day changes follows the ECB's own order. When a period carries a removal and a new value on the same day, the new value wins. This happens 34,989 times, all on 20 January and 8 September 2021, and in every case the removal is stamped one second before the new value, so the rule keeps the row the ECB entered last.

Nothing was lost in writing the files. For every series the latest workbook and one earlier one were read back, 556 in all, and every date and value agreed with the rebuilt vintage, the values to the 15 significant digits Excel stores.

The zip files also hold 44 series that the web service does not have, most of them for the United States or Japan, and those can only be taken from the zip files. None of these checks shows that the ECB kept its series up to date.

3 How Current and Complete the Data Is

There is a vintage for each of the 261 dates, but not every series in it is complete or current. Nothing in the download marks the three cases below.

Series that start late. The first vintage holds 183 of the 278 series. Another 45 start later in 2001, 26 in 2002 or 2003 and 24 in 2012, and a series has no workbook before it starts. Nine series are also missing for years. The five government bond yields and the two raw material prices have no data in any vintage from 6 April 2011 to 6 March 2024, and producer prices for durable and non-durable consumer goods have none from 21 January 2015 to 6 March 2024.

Series that have fallen behind. This report counts a series as behind when its last observation is older than was usual for it in 2015–2022, by at least three months for a monthly series, two quarters for a quarterly one or a year for an annual one, or when the ECB has left it unchanged for longer than it ever did in those years. In the most recent vintage, 9 September 2026, 107 of the 278 series are behind by this rule, most having stopped during 2025. The thresholds are a judgement, but the count does not depend on them. Halving or doubling the thresholds leaves the same 107 series behind, since the monthly ones among them are at least ten months later than usual and only one other series is late at all, by a month. Industrial production, for example, was last updated in the vintage of 29 October 2025, where it ends in August 2025. Each of the six vintages since then still ends in August 2025, although Eurostat has published figures up to July 2026. The 107 series should therefore be used only up to the vintage at which each stopped. The other 171 are up to date now, but 46 of them were behind or empty for 2 to 25 vintages in a row in 2022–2025 before catching up, including all the employment and exchange rate series, and should not be used for those vintages. That makes 153 series behind or empty in two or more consecutive vintages since 2015. The appendix gives the latest position by indicator.

Series the ECB stopped publishing. The log does not record that the ECB has stopped publishing a series, and keeps it until the ECB removes it, which can be years later. A rebuilt vintage can therefore hold values that the ECB's own file for that month leaves out. Step 5 found 296 such cases in the 2001–2014 files, most of them the five government bond yields from February 2008 to March 2011. From 2015 there are no files to show it, and the only sign is that the series falls behind. The check above flags 258 of the 296, and the same five bond yields have been unchanged again since the vintage of 11 December 2024.

In short, a series can be used as the data available on a given date unless it is behind or empty in that vintage or is one of the 296 cases. The file `series_last_update.csv` says which series are behind in the latest vintage and since when, `series_gaps.csv` lists every vintage in which a series is behind or empty, and `old_file_check.csv` lists the 296 cases. All three are in the `output` folder of the repository named in Step 4.

4 Easy Mistakes for Users

1. The download is a log of revisions, not a set of vintages.

The page says its example address “can be adapted to extract the history for any RTD series.” The file it returns is not a copy of the series at each vintage but a list of changes: a row appears only when a value is published, revised or deleted. In the page's example series, for instance, the value for 2024 is part of 10 vintages but appears only once in the file, because it has not changed since it was first published. Each row says what happened (**ACTION**: `Replace` or `Delete`) and when (**VALID_FROM** or **VALID_TO**).

To rebuild the series as it stood on a given date, take, for each period, the most recent row dated on or before that date. If that row is a `Delete`, the period was not in the series on that date. Neither the page nor any of the [nine API help pages](#) explains this, and none of them mentions these three columns.

2. Some revisions are recorded as a deletion followed by a new value.

On two days the ECB revised series by first deleting the old observations and then entering the revised values one second later. Someone who looks only at the date sees a deletion and a new value on the same day. Because deletions are listed at the end of the file, the deletion looks like the last word, and the series wrongly disappears from that vintage. Comparing the full time stamp, not just the date, avoids it.

For example, on 20 January 2021 the file deletes every quarter of real GDP at 15:30:00 and enters a revised value for each at 15:30:01. The same was done for 212 of the 278 series on that day and for 189 on 8 September 2021.

3. Requests for many series at once lose the deletions.

Each series is requested by a key with five parts, such as frequency, country and variable. The API lets a user leave parts blank to mean “any” (it calls this “wildcarding”): filling in only the first part as quarterly, for example, returns every quarterly series in one request. This is much faster than one request per series, but the file leaves out every record of a period being removed, without saying so. The data then look as if nothing was ever removed, and values the ECB withdrew still appear in later vintages.

For example, real GDP for 1991–1994 was removed on 30 November 2005. Requested on its own, real GDP shows that removal, and the vintage of 30 November 2005 starts in 1995. Requested with a wildcard, the removal is missing, and the same vintage still starts in 1991.

The fix is to name every series in full. Requesting each series on its own returns the removals; joining two full keys with + also did, in a test with real and nominal GDP.

4. The old files and the web service name the series differently.

The 2001–2014 vintages come as three zip files (monthly, quarterly and annual), each holding one CSV file per month, with one column per series. The columns are headed by the series codes in use that month, which change over time for 225 of the 268 series, and none of them is the name the web service uses. Real GDP, for example, is column BM in the quarterly files but Q.Sø.S.G_GDPM_TO_C.E in the web service. The [technical documentation](#) lists what each column contains, but not the web-service name.

The web service does say which column each series is in, but neither the page, the API help section nor the technical documentation tells the user where to look. Each series’ description has a [COMPILATION](#) field ending with the column letter, such as “series: BM”. For 266 of the 268 series that have one, the letter is right. The two M3 annual growth series have their letters swapped, and the other 10 series have none, so I matched them by comparing their data with the columns no other series had claimed, which works, but is not an ideal solution.

5. Dates.

Each vintage is dated by the day its data were frozen, not by the day of the meeting it was prepared for. Usually that is the working day before a monetary policy decision, so the vintage for the decision of 10 September 2026 is dated 9 September. This holds for 248 of the 261 vintages, including every one since 2015. The other 13, all from 2001 to 2013, were frozen two days before a decision, on the decision day itself, or in mid-January after a decision. Because the page's footnote points to the [decision dates](#), a user looking for a vintage dated on a decision day will usually not find one.

The zip files are named by Monthly Bulletin month: 14 of the 168 hold a cut-off from the previous month ([200102](#) holds 31 January 2001).

6. Downloading the full set is slow and unreliable.

A user who wants every series has to make 278 requests, one per series, and each one is slow. A request often takes one to three minutes, and some fail with a time-out ("504") and have to be repeated.

Worse, some files arrive cut off part-way through while the server reports success. This happened to 13 of the 278 series, some of them on several tries in a row, and the cause was not found. Because the [Delete](#) rows come last, a cut file loses them first, and nothing warns the user. A complete last row is not proof, because a file can be cut exactly at the end of a row. To be safe, a user should download each file twice and keep it only if the two copies are identical, which makes a full download hard work for anyone and impractical without a script.

7. Some series are not what their labels say.

Taxes less subsidies on products, at current prices ([RTD.Q.S0.S.G_TLSP_TO_U.E](#)) is identical to nominal GDP from the vintage of 5 November 2014 to that of 4 June 2025, and within 2% of it in later vintages. Only the earlier vintages are on the scale of taxes less subsidies.

Two Euro Stoxx sector indices carry contradictory labels. The ECB's own description of what each measures says cyclical and non-cyclical goods, while their titles say consumer staples and consumer discretionary.

The adjustment code is the wrong way round for ten money and credit series. The M1, M2, M3, loans and credit series whose key says not adjusted hold the ECB's seasonally adjusted figures, and those whose key says adjusted hold the unadjusted ones.

The [TITLE](#) field is blank for 17 series and shared across 88 more, so it cannot identify a series on its own either. A user has to check the key and the data itself, not the name. Both labels, and a note on the taxes series, are recorded in the workbooks.

8. A vintage can hold data that were already out of date.

A vintage is meant to show what was known on its date, but the ECB did not keep every series updated. When a series is not updated, each later vintage simply repeats it, and nothing in the download says so. In the latest vintage 107 of the 278 series are out of date in this way. Industrial production, for example, still ends in August 2025.

The case that is hardest to notice is a series that lapsed and later caught up. Unemployment ended in January 2024 in every vintage from March 2024 to June 2025, then gained 16 months at once in July 2025. Today it looks complete, but those earlier vintages do not show what was known at the time. A user should check where a series ends in each vintage before using it. Section 3 and the appendix list the cases.

5 Suggestions

1. Explain on the page that the download is a list of changes, not a set of vintages, and how to rebuild a vintage from it, including how revisions show up as a deletion followed by a new value.
2. Better still, post one file per vintage from 2015, as for 2001–2014, so that users need not rebuild anything from the logs.
3. Say on the page which series are no longer updated and since when, or resume updating them.
4. Make requests for many series at once include the periods the ECB removed. At present such requests leave them out without warning, so users end up with values the ECB has withdrawn.
5. Make the server report an error when a file is cut off, instead of reporting success.
6. State that the address covers January 2001 onwards, correct the stated update schedule, and explain why the meeting of 23 July 2026 has no vintage.
7. Say on the page that each vintage is dated by the day its data were frozen, not by the decision date.
8. Add each series' web-service name to the technical documentation. In the web service's `COMPILATION` field, fill in the column letter for the 10 series that lack it and correct the letters of the two M3 growth series.
9. Correct the page's description of the pre-2001 zip file, which holds Excel files and only one series going back before 2001, and add the missing `Cut-off.xls`.
10. Correct the series whose labels do not match their contents, fill in the blank `TITLE` fields, and make the concept names and series titles agree for the two Euro Stoxx indices.

A What Is Up to Date in the Latest Vintage

The two lists give the position in the vintage of 9 September 2026. They refer to the euro area, with Japan and the United States at the end of each. The notes after them cover data that look up to date but need care.

Up to date

Indicator	Data run to
GDP, real and nominal, and the GDP deflator	2026-Q2
Expenditure components of GDP (private and government consumption, investment, exports, imports, inventories)	2026-Q2
Gross value added, total and by sector, and taxes less subsidies in real terms	2026-Q2
Employment, total and by sector, employees and self-employed	2026-Q2
Unemployment, rate and number of persons, by age and sex	July 2026
Unit labour costs, total and by sector	2026-Q2
Labour cost index, total, wages and non-wage costs	2026-Q2
HICP, all items and every component	July or August 2026
Residential construction prices	2026-Q2
M1, M2, M3 and the M3 growth rate, seasonally adjusted (note 3)	July 2026
Loans and credit to the private sector, seasonally adjusted (note 3)	July 2026
Exports and imports of goods, values and volumes	May or June 2026
Capital account	June 2026
Euribor, 1, 3, 6 and 12 months	August 2026
Effective exchange rates of the euro	August 2026
Stock prices, Euro Stoxx and its sectors	August 2026
Economic Sentiment Indicator, the confidence indicators for industry, services, consumers, retail and construction, and capacity utilisation	August 2026
Government balances, compensation of employees and intermediate consumption	2025
New car registrations, seasonally adjusted	June 2026
United States and Japan: US consumer prices, the S&P 500 and the Nikkei	July or August 2026

Not up to date

Indicator	Last updated in the vintage of	Data end
Taxes less subsidies on products, nominal	16 April 2025	2024-Q4
Labour cost index by sector, and its seasonally adjusted total	4 June 2025	2025-Q1
HICP excluding energy and unprocessed food	4 June 2025	May 2025
Industrial production: total excluding construction, manufacturing, construction, energy, and capital, intermediate and consumer goods	29 October 2025	August 2025
Industrial production: total including construction, total excluding construction and energy, durable and non-durable consumer goods	4 June 2025	March 2025
Producer prices: total, manufacturing, energy, and capital, intermediate and consumer goods	29 October 2025	August 2025
Producer prices: durable consumer goods	4 June 2025	March 2025
Producer prices: non-durable consumer goods	11 December 2024	October 2024
Retail sales in real terms: total, food and non-food	29 October 2025	August 2025
Retail sales, total in nominal terms	4 June 2025	March 2025
Retail sales in real terms: household equipment	6 March 2024	December 2020
Retail sales in real terms: textiles and clothing	6 March 2024	April 2022
New car registrations, adjusted for working days	15 March 2023	December 2022
M1, M2, M3 and the M3 growth rate, not seasonally adjusted, and base money (note 3)	4 June 2025	April 2025
Loans and credit to the private sector, not seasonally adjusted (note 3)	4 June 2025	April 2025
Current account and its components (goods, services, primary and secondary income)	4 June 2025	March 2025
Export and import unit value indices	4 June 2025	February 2025
Government bond yields, 2 to 10 years	11 December 2024	November 2024
Brent oil price and raw material prices	11 December 2024	November 2024
Eonia, and three-month yen and dollar Libor (note 2)	April and October 2024	December 2021 to September 2024
Individual survey questions, such as order books, stocks and expectations	4 June 2025	May 2025
Government final consumption and its breakdown, consumption of fixed capital, and sales	4 June 2025	2024
Japan: GDP, industrial production, M2 and consumer prices	4 June 2025	2025-Q1 or April 2025
Japan: unemployment rate	6 March 2024	December 2023
United States: GDP, industrial production, M2, unemployment rate and unit labour costs	4 June 2025	2025-Q1 or April 2025

Notes

- 1. Up to date now, but not throughout.** Six groups in the first list were not updated for a long stretch and then caught up in a single vintage. In the vintages below they are out of date, even though the latest vintage shows nothing wrong.

Indicator	Out of date in the vintages	Data ended in
Residential construction prices	8 June 2022 to 4 June 2025	2021-Q3
Effective exchange rates of the euro	15 March 2023 to 4 June 2025	November 2022
Employment, all series	6 March 2024 to 4 June 2025	2023-Q2
Unemployment, all series	5 June 2024 to 4 June 2025	January 2024
Government balances, compensation of employees and intermediate consumption	3 May and 14 June 2023	2021
Economic Sentiment Indicator and the confidence indicators	1 February to 14 June 2023	no values at all

- 2. Rates that no longer exist.** Eonia and the three-month yen and dollar Libor rates end because the rates themselves were abolished, in January 2022, December 2022 and September 2024. No later values exist for them anywhere.
- 3. The version matters.** Several indicators are up to date in one version and not in another. Money, loans and new car registrations are up to date only when seasonally adjusted, goods trade in values and volumes but not in unit values, and taxes less subsidies in real but not in nominal terms. For money and loans the web service's key says the opposite of what the series holds (Section 4, item 7).
- 4. Values the ECB had stopped publishing.** In the vintages from February 2008 to March 2011 the government bond yields and the raw material prices appear with values, although the ECB's own files for those months leave them blank (Step 5).